

2.4.2 Misuses of Regression for Calibration

2.4.2.1 Coefficients of Correlation and Determination

Some comments about the use or abuse of the coefficient of determination r^2 or its square root, the coefficient of correlation r , are imperative. They are often alleged to be efficient criteria to decide whether a calibration model is satisfactory or not. Using proposed notations, calculation formulas are given by Eqs. (2.36) and (2.37):

Coefficient of determination

$$r^2 = \frac{SP^2(X, Y)}{SS(X) \times SS(Y)} \quad (2.36)$$

Coefficient of correlation

$$r = \frac{SP(X, Y)}{\sqrt{SS(X) \times SS(Y)}} \quad (2.37)$$

In this case, $SS(Y)$ represents the sum of the squared deviates of Y to the mean and is calculated as in Eq. (2.12) by replacing the X values by Y . The confidence interval of r is not symmetrical and two formulas are necessary to obtain lower r_L and upper limits r_U . A prior transformation using the natural logarithm is required.

Initial transformation

$$D = \frac{1}{2} \times \ln \left(\frac{1+r}{1-r} \right)$$

Confidence interval in natural logarithm

$$L = D - \frac{z_{1-\alpha/2}}{\sqrt{I-3}}$$

$$U = D + \frac{z_{1-\alpha/2}}{\sqrt{I-3}}$$

Lower limit

$$r_L = \frac{e^{2L} - 1}{e^{2L} + 1}$$

Upper limit

$$r_U = \frac{e^{2U} - 1}{e^{2U} + 1}$$

To obtain the confidence interval, the following lines should be added after the last line 29 of the worksheet Resource A.

	A	B	C	D	E
30	Coefficient of correlation: Confidence interval				
31	Alpha (risk)	5%		Alpha (risk)	5%
32	Coeff r	0.997257	=SQRT(B27)	Coeff r	0.9973
33	Transformation	3.295231	=0.5*LN((1+B32)/(1-B32))	Transformation	3.309273
34	Coverage factor	1.960	=NORM.S.INV(1-B31/2)	Coverage factor	1.960
35	Number of points	10	=COUNT(A5:A14)	Number of points	10
36	L	2.5544	=B33-B34/SQRT(B35-3)	L	2.5685
37	U	4.0360	=B33+B34/SQRT(B35-3)	U	4.0501
38	Lower limit	0.9880	=(EXP(2*B36)-1)/(EXP(2*B36)+1)	Lower limit	0.9883
39	Upper limit	0.9994	=(EXP(2*B37)-1)/(EXP(2*B37)+1)	Upper limit	0.9994

The coefficient of determination of the linear model is $r^2 = 0.9945$ and appears in cell B27 of the Resource A worksheet, delivered by LINEST, and $r^2 = 0.9947$ for the quadratic model in cell E29. To obtain the coefficients of correlation, the square roots of the coefficients of determination are extracted in cells B32 and E32, respectively. In this case, they are positive because X and Y variables are always positively correlated. When the correlation is negative, it is compulsory to force to the negative square root. Finally, it appears that the 95% confidence intervals of the coefficient of correlation r of both models are remarkably close, almost overlapping, and equal to $[0.9880; 0.9994]$ and $[0.9883; 0.9994]$, respectively. As already stated, it is impossible to decide which model is the best. For this, some fundamental reasons are helpful.

Historically, r was introduced to measure the correlation, i.e. the proper relationship between two variables. By construction, r value is always between -1 and $+1$. When it is positive, it means that the two variables tend to increase simultaneously and are claimed “positively correlated,” and in the opposite way, they are “negatively correlated”; when r is close to 0, this means the two variables are “uncorrelated.”

A test to check whether r is statistically different from 0 can be assessed or its confidence interval can be used. If the null hypothesis, i.e. $r \neq 0$, is accepted, it is concluded that the two variables are correlated as in calibration. The coefficient of determination r^2 is always positive, and varies between 0 and 1. Expressed as a percentage, it was demonstrated it can be construed as the percentage of the variation of Y explained by the regression model. The complement $1 - r^2$ can also be used to account for what is not explained by the model. An adjusted coefficient of determination also exists weighed by the number of points used for computing.

In the case of calibration, the correlation between the response and the concentration cannot be questioned since it is the mechanism that originates the analytical method principle. Therefore, both coefficients r and r^2 must be extremely high and (fortunately) always statistically different of zero. A set of arguments show that the use of r or r^2 is not relevant to assess the adequacy of the calibration model as underlined some years ago in analytical literature [6, 7].

Cited papers and others also explain why these coefficients are not suitable for selecting the best calibration model and must be avoided. They can be used for quality control (QC) purposes and to check that no calibration measurement is susceptible to be outlier. To summarize, there is no test to substantiate the hypothesis $r = 1$ and, by construction, the confidence interval of r can never include the value 1.

Therefore, it cannot be concluded that slight variations around the r value are representative of the adequacy of the calibration model. The same conclusions are applicable to r^2 . For instance, the results compiled in Table 2.3 indicate that the coefficient of determination, r^2 cannot be used to decide whether the best calibration model is linear or quadratic.

The AIC “Akaike Information Criterion (AIC)” is less well known, but it is a statistical parameter specially developed for model selection, i.e. to compare

different possible models and decide which one best fits the data. The best-fit model, according to *AIC* is the one that explains the greatest amount of variation using the fewest possible independent variables:

Akaike information criterion

$$AIC = 2K - 2\ln(L)$$

where:

- K the number of independent variables used to build the model.
- $\ln(L)$ the estimate of the logarithm of the maximum likelihood of the model (how well the model reproduces the data).

The default value of K is always 2, so if the model uses one independent variable K will be 3, if it uses two independent variables, K will be 4, and so on. To compare models, it requires calculating the *AIC* of each model. If a model is more than 2 *AIC*-units lower than another, it is considered significantly more appropriate. Referring to Table 2.3, the best calibration model would be the second-order polynomial estimated by WLS regression. In the Section 8.1 devoted to the accuracy profile, a more empirical strategy to select the best calibration model is developed.

2.4.2.2 Definitions of Linearity

There is also some confusion about the linearity often associated with method validation. In fact, linearity can receive two different definitions:

- The ability to obtain an instrumental response Y strictly related to the concentration X by a calibration curve which is a straight line.
- The ability of inverse-predicted concentrations Z_i to be proportional to the corresponding theoretical concentration X_i .

The first definition is applicable to calibration but has no great interest for validation as far as it is possible to accurately quantify an analyte with a non-linear calibration curve and many novel methods are using such models. The second definition is rather a question of trueness (see Section 4.1.3). It is the more interesting definition applicable to the validation of a method. However, many guides still insist on checking the linearity of the calibration model. In other words, numerous guidelines consider that the function applicable to all the relations between the analytical responses and the concentrations to be used for the calibration must be able to follow a linear relationship. This is wrong, and the absence of calibration curve linearity does not mean that the method cannot accurately quantify, as demonstrated by the ELISA example in the previous chapter. If a linear calibration curve is strictly required, but a curvature is observable, there are two solutions:

- Limit the calibration range to the strictly linear part. This limit will become a boundary beyond which quantification is forbidden. If an instrumental response is obtained for a sample that exceeds the bounds of the calibration range, it is

mandatory to dilute the preparation to remain within the prescribed zone. This is very restrictive, whereas the calculation of a quadratic model is quite simple.

- The second solution is to estimate the impact of this curvature on the predicted concentrations. If the bias on calculated concentration is limited, the calibration model, even if not perfectly adjusted, could be further considered.

Curvature is frequent at high concentrations, but it can also be observed in the vicinity of the blank. According to our experience, this second type of curvature is mostly found in chromatography or with various detectors, including mass spectrometers. The exact positioning of the linearity limit is tricky although there is an iterative method of finding it. The simplest procedure described in the literature is to proceed empirically using a lack-of-fit test [8]. This implies a preliminary visual examination of the calibration curve. Therefore, the proposed method requires a critical mind for its interpretation. In any case, it is essential, as a prerequisite, to carry out a graphic presentation of the results to evaluate whether a test is necessary or not.

2.4.3 Statistical Aspects of Standard Addition Method (SAM)

In Section 1.4.1, the principles of the simple standard addition method (SAM) were presented, and the accompanying calculation was explained. Another method consists of extrapolating the with-matrix straight line to a null response, as illustrated on Figure 2.5, using unpublished personal data. At the crossing point, it is possible to compute the extrapolated concentration, noted Z^* and equal to 439 ng/mL. The Resource D worksheet called SAM (Excel) illustrates the procedure when there are several standard additions.

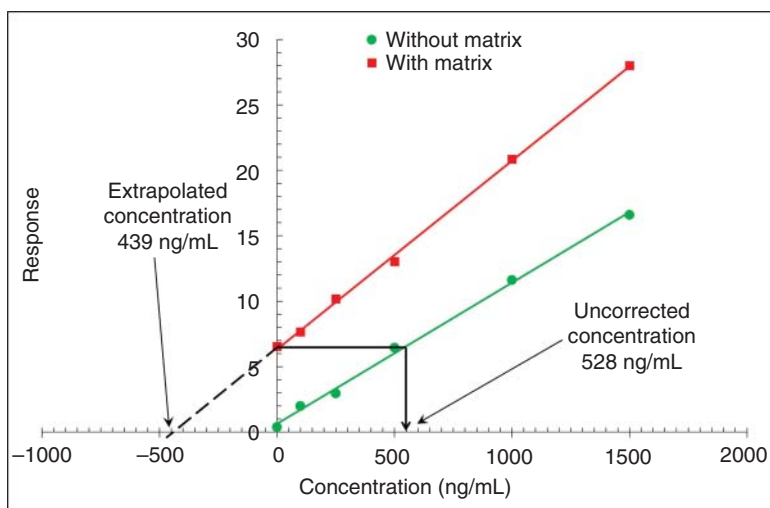


Figure 2.5 SAM – multiple point standard addition method.

This more complex design with six standard additions is recommended by some official bodies when it is necessary to demonstrate that there is no matrix effect or when using a surrogate matrix [9]. In this example, six points are measured with and without the matrix, making altogether 12 measurements. It is possible to compute with a worksheet the coefficients of the two straight lines by using the `LINEST` built-in function. Some complementary notations will help understand the rationale.

Calibration line without matrix

$$Y = a_0 + a_1X + e$$

Calibration line with authentic matrix

$$Y = b_0 + b_1X + e$$

Extrapolated concentration

$$Z^* = \frac{b_0}{b_1}$$

Uncorrected concentration

$$Z = \frac{b_0 - a_0}{a_1}$$

Recovery yield

$$RY\% = \frac{Z^*}{Z} \times 100$$

One goal of SAM is to obtain the recovery yield relevant to correcting the result of the unknown sample in presence of matrix effect, or in absence of blank matrix that can be spiked to build a representative external calibration curve. However, the procedure is also useful during the method development to check the method true-ness. For this reason, some guidelines recommend checking if both straight lines are parallel, as explained in Section 1.3, in other words if the difference between slopes is statistically equal to zero. In this specific context, an easy procedure to compare slope coefficients consists in simply comparing the confidence intervals of the slopes as explained in Section 2.2.

This simplified method is acceptable because both lines are assumed to have comparable statistical properties; otherwise, more classic statistical tests must be applied. If the confidence intervals overlap, this means that the difference is statistically not significant. In the example of Resource D, the 95% confidence intervals of a_1 and b_1 are:

Presence of an authentic matrix	95% slope confidence interval
No	[0.0100; 0.0116]
Yes	[0.0137; 0.0151]

Because intervals are not overlapping, it can be concluded that the two slopes are significantly different, with a risk of 5%, and the lines of Figure 2.5 are not parallel.